

50.007 Machine Learning, Spring 2025

Homework 3

Due 11 April 2025, 11.59pm

In this homework, we analyze the Hidden Markov Model (HMM) and its extensions. In what follows, we describe how to estimate the HMM parameters using maximum likelihood estimation (MLE) and how to decode a new observation sequence using the Viterbi algorithm. We also discuss extensions to a second-order HMM and an HMM variant that emits observation pairs.

Question 1 [20 points]

Assume that we have the following training data available for us to estimate the model parameters:

State sequence	Observation sequence
(X, X, Z, X)	(b, c, a, b)
(X, Z, Y)	(a, b, a)
(Z, Y, X, Z, Y)	(b, c, a, b, d)
(Z, Z, Y)	(c, b, a)
(X, X)	(c, a)
(Z)	(d)

The HMM is defined by:

- The initial probabilities $\pi(s)$,
- The transition probabilities $a_{u,v} = P(v | u)$,
- The emission probabilities $b_u(o) = P(o | u)$.

(a) Clearly state the parameters associated with the HMM and, using maximum likelihood estimation (MLE), compute the optimal model parameters (i.e., the initial, transition, and emission probabilities). Assume that the state space is $\{X, Y, Z\}$ and the observation alphabet is $\{a, b, c, d\}$. (Note: include transitions from the special START state and to the STOP state.) [10 points]

Answer:

We first list the transitions (including from START and to STOP) for each training instance. Note that the correct counting for transitions from state X must include the transition in instance 3, where X appears in the third position.

• **From START:**

- Instances 1, 2, and 5 start with $X \Rightarrow \text{START} \rightarrow X$: Count = 3.
- Instances 3, 4, and 6 start with $Z \Rightarrow \text{START} \rightarrow Z$: Count = 3.

- $\text{START} \rightarrow Y$: Count = 0.

Total from START = 6.

• **From X :**

- Instance 1 (X, X, Z, X):
 - * $X \rightarrow X$: 1,
 - * $X \rightarrow Z$: 1,
 - * $X \rightarrow \text{STOP}$: 1.
- Instance 2 (X, Z, Y):
 - * $X \rightarrow Z$: 1.
- **Instance 3 (Z, Y, X, Z, Y):**
 - * $X \rightarrow Z$: 1.
- Instance 5 (X, X):
 - * $X \rightarrow X$: 1,
 - * $X \rightarrow \text{STOP}$: 1.

Totals from X :

$$X \rightarrow X : 2, \quad X \rightarrow Z : 1 + 1 + 1 = 3, \quad X \rightarrow \text{STOP} : 2, \quad X \rightarrow Y : 0.$$

Total transitions from $X = 2 + 3 + 2 = 7$.

• **From Y :**

- Instance 2 (X, Z, Y): $Y \rightarrow \text{STOP}$: 1.
- Instance 3 (Z, Y, X, Z, Y): $Y \rightarrow X$: 1 and $Y \rightarrow \text{STOP}$: 1.
- Instance 4 (Z, Z, Y): $Y \rightarrow \text{STOP}$: 1.

Totals from Y : $Y \rightarrow X$: 1, $Y \rightarrow \text{STOP}$: 3, and no transitions to Y or Z . (Total = 4)

• **From Z :**

- Instance 1 (X, X, Z, X): $Z \rightarrow X$: 1.
- Instance 2 (X, Z, Y): $Z \rightarrow Y$: 1.
- Instance 3 (Z, Y, X, Z, Y):
 - * First Z : $Z \rightarrow Y$: 1,
 - * Second Z : $Z \rightarrow Y$: 1.
- Instance 4 (Z, Z, Y):
 - * First Z : $Z \rightarrow Z$: 1,
 - * Second Z : $Z \rightarrow Y$: 1.
- Instance 6 (Z): $Z \rightarrow \text{STOP}$: 1.

Totals from Z :

$$Z \rightarrow X : 1, \quad Z \rightarrow Y : 4, \quad Z \rightarrow Z : 1, \quad Z \rightarrow STOP : 1.$$

Total = 7.

Thus, the MLE transition probabilities are:

$$\begin{aligned} a_{\text{START},X} &= \frac{3}{6} = \frac{1}{2}, & a_{\text{START},Z} &= \frac{3}{6} = \frac{1}{2}, \\ a_{\text{START},Y} &= 0. \\ a_{X,X} &= \frac{2}{7}, & a_{X,Z} &= \frac{3}{7}, \\ a_{X,Y} &= 0, & a_{X,STOP} &= \frac{2}{7}. \\ a_{Y,X} &= \frac{1}{4}, & a_{Y,STOP} &= \frac{3}{4}, & a_{Y,Y} &= a_{Y,Z} = 0. \\ a_{Z,X} &= \frac{1}{7}, & a_{Z,Y} &= \frac{4}{7}, \\ a_{Z,Z} &= \frac{1}{7}, & a_{Z,STOP} &= \frac{1}{7}. \end{aligned}$$

Emission Counts:

- **For state X :**

- Instance 1: Emits $b, c, b \Rightarrow$ counts: $b=2, c=1$.
- Instance 2: Emits $a \Rightarrow$ count: $a=1$.
- Instance 3: Emits $a \Rightarrow$ count: $a=1$.
- Instance 5: Emits $c, a \Rightarrow$ counts: $c=1, a=1$.

Totals for X : $a=3, b=2, c=2, d=0$ (Total = 7).

- **For state Y :**

- Instance 2: Emits $a \Rightarrow$ count: $a=1$.
- Instance 3: Emits $c, d \Rightarrow$ counts: $c=1, d=1$.
- Instance 4: Emits $a \Rightarrow$ count: $a=1$.

Totals for Y : $a=2, c=1, d=1, b=0$ (Total = 4).

- **For state Z :**

- Instance 1: Emits $a \Rightarrow$ count: $a=1$.
- Instance 2: Emits $b \Rightarrow$ count: $b=1$.
- Instance 3: Emits $b, b \Rightarrow$ count: $b=2$.
- Instance 4: Emits $c, b \Rightarrow$ counts: $c=1, b=1$.
- Instance 6: Emits $d \Rightarrow$ count: $d=1$.

Totals for Z : $a=1, b=4, c=1, d=1$ (Total = 7).

Thus, the MLE emission probabilities are:

$$\begin{aligned} b_X(a) &= \frac{3}{7}, & b_X(b) &= \frac{2}{7}, \\ b_X(c) &= \frac{2}{7}, & b_X(d) &= 0; \\ b_Y(a) &= \frac{2}{4} = \frac{1}{2}, & b_Y(b) &= 0, \\ b_Y(c) &= \frac{1}{4}, & b_Y(d) &= \frac{1}{4}; \\ b_Z(a) &= \frac{1}{7}, & b_Z(b) &= \frac{4}{7}, \\ b_Z(c) &= \frac{1}{7}, & b_Z(d) &= \frac{1}{7}. \end{aligned}$$

(b) Smoothing Extension.

To address zero counts, we use Laplace (add- α) smoothing. For transitions:

$$\hat{a}_{u,v} = \frac{\text{Count}(u \rightarrow v) + \alpha}{\text{Count}(u) + \alpha \cdot K},$$

where K is the number of possible transitions from state u (including STOP) and $\alpha > 0$ (typically $\alpha = 1$). For emissions:

$$\hat{b}_u(o) = \frac{\text{Count}(u \rightarrow o) + \alpha}{\text{Count}(u) + \alpha \cdot |\mathcal{O}|},$$

with $|\mathcal{O}| = 4$.

Question 2 [10 points]

Now, consider during the evaluation phase, you are given the following new observation sequence. Using the (unsmoothed) MLE parameters from Question 1, find the most probable state sequence for the observation sequence (a, b) using the Viterbi algorithm.

State sequence	Observation sequence
(?, ?)	(a, b)

Answer:

Let $\pi(k, s)$ denote the highest probability of any state sequence ending in state s at time step k , with $\pi(0, \text{START}) = 1$ and $\pi(0, s) = 0$ for $s \neq \text{START}$.

Time Step 1 (Observation: a)

- For state X :

$$\pi(1, X) = \pi(0, \text{START}) \cdot a_{\text{START}, X} \cdot b_X(a) = 1 \cdot \frac{1}{2} \cdot \frac{3}{7} = \frac{3}{14}.$$

- For state Z :

$$\pi(1, Z) = \pi(0, \text{START}) \cdot a_{\text{START}, Z} \cdot b_Z(a) = 1 \cdot \frac{1}{2} \cdot \frac{1}{7} = \frac{1}{14}.$$

- For state Y : Since $a_{\text{START}, Y} = 0$, $\pi(1, Y) = 0$.

Time Step 2 (Observation: $\mathbf{b}$)

For each candidate state at time 2, consider all possible predecessors from time 1.

- **For state X :**

- From X :

$$\pi(1, X) \cdot a_{X,X} \cdot b_X(b) = \frac{3}{14} \cdot \frac{2}{7} \cdot \frac{2}{7} = \frac{12}{686} = \frac{6}{343}.$$

- From Z :

$$\pi(1, Z) \cdot a_{Z,X} \cdot b_X(b) = \frac{1}{14} \cdot \frac{1}{7} \cdot \frac{2}{7} = \frac{2}{686} = \frac{1}{343}.$$

Therefore, $\pi(2, X) = \frac{6}{343}$ (with backpointer from X).

- **For state Z :**

- From X :

$$\pi(1, X) \cdot a_{X,Z} \cdot b_Z(b) = \frac{3}{14} \cdot \frac{3}{7} \cdot \frac{4}{7} = \frac{36}{686} = \frac{18}{343}.$$

- From Z :

$$\pi(1, Z) \cdot a_{Z,Z} \cdot b_Z(b) = \frac{1}{14} \cdot \frac{1}{7} \cdot \frac{4}{7} = \frac{4}{686} = \frac{2}{343}.$$

Thus, $\pi(2, Z) = \frac{18}{343}$ (with backpointer from X).

- **For state Y :** Since $b_Y(b) = 0$, we have $\pi(2, Y) = 0$.

Termination:

Transition to the STOP state is computed as:

$$\pi(3, \text{STOP}) = \max \left\{ \pi(2, X) \cdot a_{X, \text{STOP}}, \quad \pi(2, Z) \cdot a_{Z, \text{STOP}}, \quad \pi(2, Y) \cdot a_{Y, \text{STOP}} \right\}.$$

Substituting the values:

- From X : $\frac{6}{343} \cdot \frac{2}{7} = \frac{12}{2401}.$

- From Z : $\frac{18}{343} \cdot \frac{1}{7} = \frac{18}{2401}.$

- From Y : 0.

Since $\frac{18}{2401} > \frac{12}{2401}$, the best final probability is obtained via state Z . Tracing back, the backpointer for state Z at time 2 points to state X at time 1. Thus, the optimal state sequence is **($\mathbf{X}, \mathbf{Z}$)**.

Question 3 [20 points]

The HMM can be extended to a second-order model where

$$p(x_1, \dots, x_n, y_1, \dots, y_n) = \prod_{i=1}^{n+1} p(y_i \mid y_{i-2}, y_{i-1}) \cdot \prod_{i=1}^n p(x_i \mid y_i),$$

with $y_{-1} = y_0 = \text{START}$ and $y_{n+1} = \text{STOP}$.

(a) Initialization and State Representation [4 points]

Answer: Define the state pair $\langle u, v \rangle$, where u is the state at time $i - 1$ and v at time i . Initialize:

$$\pi(0, \langle \text{START}, \text{START} \rangle) = 1, \quad \pi(0, \langle u, v \rangle) = 0 \quad \text{for } \langle u, v \rangle \neq \langle \text{START}, \text{START} \rangle.$$

For $i = 1$, set:

$$\pi(1, \langle \text{START}, v \rangle) = a_{\text{START}, \text{START}, v} \cdot b_v(x_1) \quad \forall v.$$

(b) Recursion [8 points]

Answer: For $k \geq 2$, update:

$$\pi(k, \langle u, v \rangle) = \max_{t \in \{X, Y, Z\}} \left\{ \pi(k-1, \langle t, u \rangle) \cdot a_{t, u, v} \cdot b_v(x_k) \right\}.$$

(c) Termination and Backtracking [8 points]

Answer: Terminate by transitioning to STOP:

$$\pi(n+1, \langle u, \text{STOP} \rangle) = \max_t \left\{ \pi(n, \langle t, u \rangle) \cdot a_{t, u, \text{STOP}} \right\}.$$

Backtracking is performed by storing the argmax choices at each step. Alternatively, one may recover the final two states via:

$$\langle y_{n-1}^*, y_n^* \rangle = \arg \max_{u, v} \left\{ \pi(n, \langle u, v \rangle) \cdot a_{u, v, \text{STOP}} \right\}.$$

Question 4 [20 points]

Now consider an extended graphical model where each state Z emits an observation pair (X, Y) , and the Y -component is generated conditioned on both Z and X .

Assume a predefined set of states $\{0, 1, \dots, N-1, N\}$ with $0 = \text{START}$ and $N = \text{STOP}$. Define

$$c_u(x, y) = P(y \mid u, x).$$

Then, analogous to the standard HMM, the forward score at time i for state u is:

$$\alpha_u(i) = P(x_1, y_1, \dots, x_i, y_i, z_i = u).$$

Forward Algorithm:

- **Base Case:** For $i = 1$,

$$\alpha_u(1) = a_{\text{START}, u} \cdot b_u(x_1) \cdot c_u(x_1, y_1), \quad \forall u \in \{1, \dots, N-1\}.$$

- **Recursion:** For $i = 2, \dots, n$,

$$\alpha_u(i) = \sum_v \alpha_v(i-1) \cdot a_{v,u} \cdot b_u(x_i) \cdot c_u(x_i, y_i),$$

where the sum is over all possible previous states v .

Backward Algorithm:

- **Base Case:** For $i = n$,

$$\beta_u(n) = a_{u,\text{STOP}} \cdot b_u(x_n) \cdot c_u(x_n, y_n), \quad \forall u \in \{1, \dots, N-1\}.$$

- **Recursion:** For $i = n-1, \dots, 1$,

$$\beta_u(i) = \sum_v a_{u,v} \cdot b_v(x_{i+1}) \cdot c_v(x_{i+1}, y_{i+1}) \cdot \beta_v(i+1),$$

summing over all possible next states v .

Time Complexity: At each of the n time steps, for each of the N states we perform a sum over N states, yielding an overall complexity of $O(nN^2)$.